

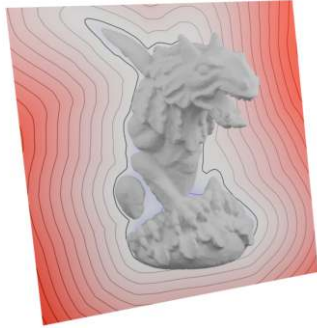
Internship proposal

Fitting surface meshes on implicit distance fields

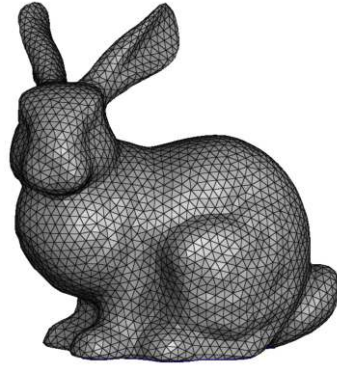
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The CRAFT team is an Inria project that aims at developing methods for designing and modeling deformable objects by combining geometry, physics, and numerical optimization. Our applications call for the setup and the resolution of inverse problems involving geometrical objects that needs to deform according to physical forces and material constraints. For these deformations to be plausible, the data structure representing the object has to be carefully chosen and preserved for the computation to make sense.

In this internship, we will focus on deformation and evolution of an object so that it conforms to a target surface. The (fixed) target will be represented by its implicit signed distance field encoded in a neural network (Figure 1a), which will allow for automatic differentiation and quick evaluation of the distance and its gradient. On the other hand, the deformable object will be a polygonal mesh (Figure 1b) [1]. The mesh is indeed the most widespread data structure used in this context, yet its discretization of the surface into polygonal (often triangular) faces poses a challenge in terms of validity: as faces can numerically degenerate, flip or overlap, the represented object does not make sense physically anymore. As such, the fitting objective of variable surface to the target needs to be modulated by a validity condition, which takes the form of regular remeshing (as in *shrinkwrapping* algorithms [1, 2]), distortion energies [5] or hard-constraints over the available degrees of freedom [3]. By considering the elastic properties of the material, our hope is to regularize the fitting energy to achieve a plausible surface fitting.



(a) Neural distance field of a gargoyle computing with the method of Coiffier and Béthune [4].



(b) Triangle mesh of the *Stanford Bunny*.

We will consider applications to elastic simulation for design, as well as more general geometry processing problems such as remeshing algorithms and surface extraction from implicit surfaces.

Required skills

The intern will have to be comfortable with either python or C++ for implementation. They should have some knowledge about geometry processing data structure and algorithms, as well as numerical optimization algorithms.

Practical information

The internship will take place in the CRAFT research team in the Inria center of the Grenoble Alpes University (655 Avenue de l'Europe, 38330 Montbonnot-Saint-Martin, France). It can last from 4 to 6 months.

References

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- [3] Guillaume Coiffier, Justine Basselin, Nicolas Ray, and Dmitry Sokolov. 2021. Parametric Surface Fitting on Airborne Lidar Point Clouds for Building Reconstruction. *Computer-Aided Design* 140 (Nov. 2021), 103090. <https://doi.org/10.1016/j.cad.2021.103090>
- [4] Guillaume Coiffier and Louis Béthune. 2024. 1-Lipschitz Neural Distance Fields. *Computer Graphics Forum* 43, 5 (2024), e15128. <https://doi.org/10.1111/cgf.15128>
- [5] Vladimir Garanzha, Igor Kaporin, Liudmila Kudryavtseva, François Protais, Nicolas Ray, and Dmitry Sokolov. 2021. Foldover-Free Maps in 50 Lines of Code. *ACM Transactions on Graphics* 40, 4 (Aug. 2021), 1–16. <https://doi.org/10.1145/3450626.3459847>
- [6] Vincent Nivolières, Dong-Ming Yan, and Bruno Lévy. 2014. Fitting Polynomial Surfaces to Triangular Meshes with Voronoi Squared Distance Minimization. *Engineering with Computers* 30, 3 (July 2014), 289–300. <https://doi.org/10.1007/s00366-012-0291-9>
- [7] François Protais, Gianmarco Cherchi, and Marco Livesu. [n. d.]. Versatile Volume Fitting with Automatic Feature Preservation. ([n. d.]).